



AI Engineer Roadmap 2026

From Zero to Job-Ready in 10 Months

LLMs • RAG • AI Agents • MCP • Fine-Tuning • Multimodal

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PHASE 1

Foundation

Weeks 1-4



Python Mastery

- Python basics — variables, loops, functions, OOP
- Libraries: NumPy, Pandas, Matplotlib
- Jupyter Notebooks for experimentation
- Virtual environments, pip, package management
- Resource: Python.org docs, freeCodeCamp Python course

FREE



Data Fundamentals

- Data cleaning & preprocessing with Pandas
- Data visualization (Matplotlib, Seaborn, Plotly)
- SQL basics — SELECT, JOIN, GROUP BY
- CSV, JSON, API data handling
- Resource: Kaggle Learn, W3Schools SQL

FREE

PHASE 2

Machine Learning Core

Weeks 5-10



Math (Just Enough)

- Linear Algebra — vectors, matrices, dot products

- Statistics — mean, median, standard deviation, distributions
- Probability — Bayes theorem, conditional probability
- Calculus — gradients (just the concept)
- Resource: 3Blue1Brown YouTube, Khan Academy **FREE**

Classical ML

- **Supervised**: Linear/Logistic Regression, Decision Trees, Random Forest, SVM
- **Unsupervised**: K-Means, PCA, DBSCAN
- Model evaluation: accuracy, precision, recall, F1, confusion matrix
- Feature engineering, train/test split, cross-validation
- **Scikit-learn** library mastery
- Resource: Andrew Ng's ML course (Coursera) **FREE**

PROJECT

Build a salary predictor or spam classifier using Scikit-learn

PHASE 3 Deep Learning

Weeks 11-16

Neural Networks

- Perceptrons, activation functions, backpropagation
- CNNs (Convolutional Neural Networks) — image tasks
- RNNs, LSTMs — sequence data
- Framework: **PyTorch** (industry standard 2026)
- Resource: Andrej Karpathy's Neural Networks Zero to Hero **FREE**

Transformers & Attention

- Self-attention mechanism — the core of modern AI
- Transformer architecture (foundation of GPT, Claude, Gemini)
- BERT, GPT architecture understanding
- **Hugging Face Transformers** library
- Resource: "Attention Is All You Need" paper + Jay Alammar's blog **FREE**

PROJECT

Fine-tune a sentiment analysis model on Hugging Face

PHASE 4

LLMs & Prompt Engineering

Weeks 17-20

**Large Language Models**

HOT

- How LLMs work — tokenization, context windows, temperature, top-p
- OpenAI AP (GPT-4o, GPT-5)
- Anthropic Claude AP (Claude Opus, Sonnet)
- Google Gemini API
- Open-source: Llama 3.3, Mistral, Qwen 2.5, DeepSeek
- Running local models: Ollama, vLLM, llama.cpp

**Prompt Engineering**

- Zero-shot, few-shot, chain-of-thought prompting
- System prompts, role-based prompting
- Structured outputs (JSON mode)
- Prompt chaining & decomposition
- Adversarial prompting & safety

**PROJECT**

Build a CLI tool that uses Claude/GPT to analyze and review code

PHASE 5

RAG & Vector Databases

Weeks 21-24

**Retrieval-Augmented Generation**

HOT

- What is RAG — making LLMs talk to YOUR data
- Document loading — PDFs, websites, databases
- Text chunking strategies (fixed, semantic, recursive)
- Embedding models: OpenAI text-embedding-3, Cohere, e5, bge
- Vector databases: Pinecone, ChromaDB, Weaviate, Qdrant
- Retrieval: similarity search, MMR, hybrid search
- Re-ranking: Cohere Rerank, cross-encoders
- Evaluation: faithfulness, relevance, context recall

**RAG Frameworks**

- LangChain — most popular, huge ecosystem
- LlamaIndex — data-focused, great for complex retrieval

- Haystack — production-ready pipelines

PROJECT

Build a "Chat with your PDF" app — upload any document, ask questions, get answers with page citations

PHASE 6 AI Agents & Agentic AI

Weeks 25-30



AI Agents HOTTEST SKILL 2026

- What are AI agents — reasoning + acting (ReAct pattern)
- Tool use — giving LLMs ability to call functions, APIs, databases
- Planning — multi-step task decomposition
- Memory — short-term (conversation) + long-term (vector DB)
- Frameworks: LangGraph, CrewAI, AutoGen, OpenAI Agents SDK



MCP — Model Context Protocol NEW 2026

- Open standard by Anthropic for connecting AI to external tools
- Build MCP servers (expose your tools to any AI)
- Build MCP clients (connect AI to any tool)
- Used in Claude Desktop, Claude Code, Cursor
- Becoming the USB standard for AI — learn it NOW



Multi-Agent Systems

- Agent orchestration — manager agents delegating to specialists
- Agent communication protocols
- Human-in-the-loop workflows
- Error handling & fallback strategies

PROJECT 1

Build an MCP server that connects AI to your database

PROJECT 2

Build a research agent that searches the web, summarizes findings, and writes a report

PHASE 7 Fine-Tuning & Multimodal

Weeks 31-34

Fine-Tuning LLMs

- When to fine-tune vs RAG vs prompt engineering
- LoRA & QLoRA — efficient fine-tuning (no expensive GPUs needed)
- Dataset preparation — instruction-response pairs
- Training: Hugging Face TRL, Axolotl
- Models: Llama 3.3 8B, Mistral 7B, Qwen 2.5
- Evaluation: benchmarks, human eval, LLM-as-judge

Multimodal AI NEW 2026

- Vision-Language: GPT-4o, Claude Vision, Gemini
- Image generation: Stable Diffusion, DALL-E 3, Flux
- Audio: Whisper (transcription), ElevenLabs (TTS)
- Video understanding: Gemini 1.5 Pro

PROJECT

Fine-tune Llama 3.3 8B on 1,000 custom instruction pairs using LoRA

PHASE 8 Production & Deployment

Weeks 35-38

MLOps & AI in Production

- API development: FastAPI, Flask
- Containerization: Docker
- Deployment: Vercel, Railway, AWS Lambda, GCP Cloud Run
- Monitoring: LangSmith, Langfuse, Weights & Biases
- Cost optimization — caching, model routing, token management
- Guardrails — content filtering, output validation

AI Safety & Ethics

- Hallucination detection & mitigation
- Prompt injection attacks & defenses
- Data privacy — PII handling
- Red teaming your own systems

PHASE 9

Portfolio Projects

Weeks 39-42

 Build These 5 Projects

- 1. RAG Chatbot — Chat with any PDF/website with cited answers
- 2. AI Agent with MCP — Multi-tool agent connected to real APIs
- 3. Fine-Tuned Model — Custom model for a specific domain
- 4. Full-Stack AI App — Next.js + AI backend, deployed with auth & payments
- 5. Open Source Contribution — Contribute to LangChain, LlamaIndex, or any AI framework

 Tool Stack 2026

LANGUAGE Python, TypeScript	LLM APIS OpenAI, Claude, Gemini, Groq
OPEN MODELS Llama 3.3, Mistral, Qwen, DeepSeek	LOCAL INFERENCE Ollama, vLLM, llama.cpp
RAG LangChain, LlamaIndex	VECTOR DB Pinecone, ChromaDB, Weaviate
AGENTS LangGraph, CrewAI, AutoGen	INTEGRATION MCP (Model Context Protocol)
FINE-TUNING Hugging Face TRL, LoRA, Axolotl	DEPLOYMENT FastAPI, Docker, Vercel, AWS
MONITORING LangSmith, Langfuse	FRONTEND Next.js, Streamlit, Gradio

 Salary Range (India 2026)

LEVEL	EXPERIENCE	SALARY
Junior AI Engineer	0-2 years	₹8-15 LPA
Mid AI Engineer	2-4 years	₹15-30 LPA
Senior AI Engineer	4-7 years	₹30-55 LPA

Lead / Staff	7+ years	₹55-90+ LPA
Remote (US companies)	3+ years	\$80K-180K



Complete Timeline

PHASE	DURATION	FOCUS
Foundation	4 weeks	Python + Data
ML Core	6 weeks	Classical ML + Math
Deep Learning	6 weeks	Neural Networks + Transformers
LLMs	4 weeks	APIs + Prompt Engineering
RAG	4 weeks	Vector DBs + Retrieval
Agents	6 weeks	Agentic AI + MCP
Fine-Tuning	4 weeks	LoRA + Multimodal
Production	4 weeks	Deployment + MLOps
Portfolio	4 weeks	5 Projects
Total: ~10 months (can compress to 4-5 months if full-time)		



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